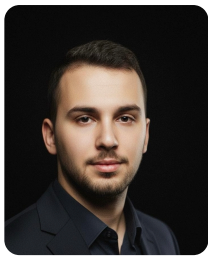
Muhammet Utku Küçük

Backend Developer

Fatih, Istanbul, Türkiye | +90 554 006 75 23 | utku@utkukucuk.com | utkukucuk.com | linkedin.com/in/utkukucuk | github.com/utku-kucuk



SUMMARY

Backend developer with a B.Sc. in Computer Engineering (2025) and hands-on experience building REST APIs, backend services and admin panels with PHP (Laravel), Node.js/TypeScript and C#, gained through a remote backend contract at Covalent, corporate client projects and two software internships, plus Python/OpenCV work in a university UAV R&D team. Comfortable with Docker, Linux, Nginx, PostgreSQL, MySQL, MSSQL, MongoDB and GitHub Actions CI/CD, writing documented, secured APIs (OpenAPI, JWT, RBAC), and turning data into reports with SQL and Python (Pandas). Now targeting **Java/Spring Boot** backend roles, with Kubernetes and deeper AWS experience as the next focus.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, PHP, C#, SQL, C/C++

Backend: Spring Boot, Spring Data JPA, Hibernate, Node.js, Express.js, Laravel, ASP.NET, REST APIs, Microservices, OpenAPI/Swagger, JWT, RBAC, LDAP/Active Directory

Databases: PostgreSQL, MySQL, MSSQL, MongoDB, SQLite, Redis

Data Analysis: SQL (joins, aggregations, window functions), Python (Pandas, NumPy), Matplotlib, Jupyter Notebook, Excel (pivot tables), Power BI, ETL, data cleaning, data modeling, data visualization, reporting

DevOps & Cloud: Docker, Docker Compose, Linux, Nginx (reverse proxy, SSL/TLS), GitHub Actions (CI/CD), AWS (EC2, S3), Cloudflare

Testing & Tools: Unit Testing (JUnit, Jest, PHPUnit), Postman, Git, Maven, Jira, Agile/Scrum, Code Review

Frontend & IoT: React, Astro, Tailwind CSS, Bootstrap, jQuery; ESP32, ESPHome, MQTT, Home Assistant

EXPERIENCE

R&D Project Evaluator | Zaim Teknopark · Freelance · Remote — **Oct 2026 – Present**

- Evaluate R&D proposals and progress reports of software, cloud and IoT projects for technical feasibility, architecture, innovation and R&D eligibility, and recommend approval or revision.

Backend Developer | Covalent · Contract · Remote — **Sep 2025 – Feb 2026**

- Built and maintained Node.js/TypeScript backend services and REST endpoints for the Covalent Unified API, serving on-chain data (token balances, transactions, NFT metadata) to client applications.
- Developed backend features for a token/NFT marketplace dApp: listing and transaction-history endpoints, wallet-based sign-in, input validation and PostgreSQL data models.
- Added Redis response caching, pagination and rate-limit handling with retries for upstream calls; documented endpoints with OpenAPI, covered them with Jest unit tests and shipped through PR-based code reviews in an Agile team.

Software Developer (Computer Vision) | Nova UAV Team · University R&D Team · Istanbul — **Nov 2024 – Apr 2025**

- Worked on the team's Python and OpenCV image-processing modules for UAV camera feeds (preprocessing, color segmentation, object detection) and helped test and tune them on onboard hardware.

Software Developer (Backend) | Teknolojik Çözüm · Full-time, then part-time during studies · Istanbul — **Aug 2022 – Apr 2023**

- Developed backend features, REST APIs and MySQL-backed admin panels with PHP (Laravel) and JavaScript (jQuery) for corporate client projects, including schema design and SQL reports.
- Containerized applications with Docker and Docker Compose and configured Nginx reverse proxies with SSL/TLS on Linux servers.
- Set up CI/CD pipelines with GitHub Actions and handled production troubleshooting through log analysis and root-cause investigation.

Software Engineering Intern | BİL Holding · Istanbul — Jul 2021 – Aug 2021

- Developed REST API endpoints and business logic in PHP (Laravel) for internal web applications.
- Built responsive pages and forms with HTML, CSS, JavaScript (jQuery) and Bootstrap and connected them to the backend APIs.
- Wrote MySQL queries, designed tables and built reporting screens and Excel exports for internal data.
- Fixed bugs, wrote PHPUnit tests, took part in code reviews with Git and wrote technical documentation.

Software Development Intern | Kale Kalıp · Istanbul — Jul 2019 – Aug 2019

- Added modules and features to internal business applications in C# (.NET) backed by MSSQL.
- Wrote Python ETL scripts (Pandas, Matplotlib) to collect, clean, chart and store sensor data from production-floor IoT devices.
- Prepared SQL queries and Excel pivot-table reports on production data for the operations team.
- Assisted the IT team with server, network and end-user support.

PROJECTS

Commerce Platform: E-Commerce Backend — GitHub

Java 21, Spring Boot 3, Spring Data JPA, Hibernate, H2, Maven, Spring Boot Actuator

- E-commerce backend in active development: JPA domain model (users, addresses, roles, products, order status lifecycle) with audit timestamps and Actuator health checks; REST APIs, JWT security and Docker Compose come next.

Viducate: Video Learning Platform — GitHub · Live demo

Node.js, Express.js, PostgreSQL, Sequelize (ORM), FFmpeg, JWT, LDAP, Nginx, PM2, React

- Video courses, quizzes and progress tracking for three roles (Admin, Teacher, Student): chunked lecture uploads, HTTP byte-range streaming and FFmpeg transcoding to multiple resolutions with duration extraction.
- JWT authentication with LDAP/Active Directory sign-in, role-based access, CSV bulk user import and rate limiting; deployed on Linux with PM2 behind an Nginx reverse proxy.

IronEcho: Offline-First Fitness Platform — GitHub

Node.js, Express.js, MongoDB, Socket.io, JWT, Joi, React Native, TypeScript, Docker Compose

- Self-hosted Express REST API with JWT access/refresh tokens, Joi validation and per-user real-time WebSocket updates (Socket.io); the React Native client queues offline writes and syncs them with last-write-wins conflict resolution; deployed with Docker Compose.

HireHub: Role-Based Job Board — GitHub · Live demo

React 19, React Router 7, Tailwind CSS 4, Vite, GitHub Actions

- Job board for job seekers, employers and admins with role-based route guards and a REST-shaped service layer over a mocked API; linted, built and deployed by GitHub Actions.

utkukucuk.com: Personal Portfolio — Website

Astro, TypeScript, Tailwind CSS, WebGL, GitHub Actions, Lighthouse CI, Cloudflare

- Bilingual (EN/TR) site with a WebGL hero and strict CSP; every pull request passes lint, type-check and Lighthouse CI.

EDUCATION

B.Sc. in Computer Engineering (English) | Istanbul Aydın University — 2022 – 2025

Started at Bahçeşehir University, Computer Engineering (2020 – 2022); completed via lateral transfer.

Associate Degree in Computer Programming | Istanbul Aydın University — 2018 – 2020

Information Technologies (Software & Networking) | Avcılar Anatolian Technical High School — 2014 – 2018

Vocational training in programming and computer networks; IT internships at Kale Holding and Boğaziçi Döküman Çözümleri.

LANGUAGES

Turkish (Native) · English (B2 – Professional Working Proficiency)